

CSA15 - CLOUD COMPUTING AND BIG DATA ANALYTICS

INDIVIDUAL CONTRIBUTION AND REFLECTION

Field	Student Entry
Slot	B
Assignment Title	Cloud-Ready Smart Campus Service Platform: API Implementation, Workload Sizing and Virtualized Deployment
Team Number	6
Date of Submission	02-09-2026
GitHub Repository	https://github.com/tharu-ui/CSA1512-Group-Assessment-Repo.git

TYPE DIRECTLY IN THIS DOCUMENT. Replace all bracketed placeholders with original content. Do not reproduce the assignment question, detailed instructions or another team's text.

Field	Student Entry
Student Name	D.S.THARUNIYA PRIYA
Register Number	192511317
Team Number	6
Assigned Module / Responsibility	Auth & Access Control
Team GitHub Repository	https://github.com/tharu-ui/CSA1512-Group-Assessment-Repo.git

1. Work Personally Completed

Personal Contribution

I was responsible for the Auth & Access Control module of the Smart Campus Service Platform. I worked on the authentication flow for student and staff users, including login using register number and password and generation of a JWT token after successful authentication. I also contributed to the access-control design where authenticated requests are validated before allowing users to access protected API operations.

I worked with the Express.js backend structure and helped define the authentication route and middleware-based security approach. The authentication design uses JWT tokens so that the user's identity and role can be verified for subsequent API requests. The two supported roles are student and canteen_staff, with role-based access used to restrict staff-only operations.

I also considered input validation and error handling for authentication requests. Invalid credentials or missing authentication information should be rejected instead of allowing unauthorised access. The JWT secret is intended to be stored through environment variables rather than being hard-coded in the source code.

I tested the authentication behaviour along with the protected API routes and verified that requests without valid authentication are rejected. I also contributed to the GitHub repository, documentation and testing evidence for the completed module.

Evidence Type	File / Module	Commit / URL	What It Proves
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Code	Auth & Access Control / authentication route	https://github.com/tharu-ui/CSA1512-Group-Assessment-Repo	Shows my assigned authentication implementation and repository contribution
Code	JWT authentication / middleware	https://github.com/tharu-ui/CSA1512-Group-Assessment-Repo	Shows token-based authentication and protected API access
Test	Unauthorized API access	https://github.com/tharu-ui/CSA1512-Group-Assessment-Repo/blob/main/canteen_test_evidence.png.png	Demonstrates that an unauthenticated request is rejected with HTTP 401

2. Individual Implementation and Result Evidence

Implementation Evidence

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27 packages are looking for funding
run 'npm fund' for details
found 0 vulnerabilities

D:\canteen-module>node tests/run_tests.js

=== CSA15 Canteen Module - Test Run ===
Run at: Wed Sep 02 2026 11:34:23 GMT+0530 (India Standard Time)
=====

Test 1: Crowd prediction with sufficient history (7-day, heavy traffic)
Expected: level = High, wait ~15 min
Observed status: 200
Observed body: {"slot":"12:30","level":"High","expectedWaitMins":15,"avgCount":51.2}
Result: PASS

Test 2: Crowd prediction with insufficient history (1 day of data)
Expected: falls back to Medium default
Observed status: 200
Observed body: {"slot":"12:30","level":"Medium","expectedWaitMins":8,"note":"fallback: insufficient history (1 day(s) of data)"}
Result: PASS

Test 3: Token request when slot is already at capacity (60/60)
Expected: 409 SLOT_FULL
Observed status: 409
Observed body: {"error":"SLOT_FULL"}
Result: PASS

Test 4: Unauthorized token request (missing Authorization header)
Expected: 401
Observed status: 401
Observed body: {"error":"Missing or invalid Authorization header"}
Result: PASS

4/4 tests passed.
D:\canteen-module>
  
```

Test Performed	Expected	Observed	Correction / Decision
Login with valid student/staff credentials	User should be authenticated and receive a JWT token	Authentication request returned a successful response with token and role	Accepted the JWT-based authentication flow
Login with invalid credentials	Request should be rejected	Invalid authentication was rejected	Kept authentication failure handling enabled
Protected API request without Authorization header	Request should return HTTP 401	Request returned 401 — {"error": "Missing or invalid Authorization header"}	Confirmed authentication middleware blocks unauthorised access
Access to role-protected functionality	Only the permitted role should access the operation	Requests are checked using the authenticated role	Maintained student and canteen_staff role separation

3. Technical Decision

Decision and Evidence

I personally supported the decision to use JWT-based authentication for the API instead of maintaining server-side login sessions. The main alternatives considered were session-based authentication and token-based authentication. JWT was selected because the project is designed as a cloud-ready REST API and stateless authentication is more suitable for APIs that may later be deployed across multiple instances.

With JWT, the server can validate the token included in the Authorization header without maintaining a separate session for every logged-in user. The token can also carry the user's role, allowing the API middleware to distinguish between student and canteen_staff users. This fits the project's requirement for authenticated access and role-based control.

The decision was supported by testing protected API requests. A request without a valid Authorization header was rejected with HTTP 401, showing that the authentication mechanism prevents unauthorised access before the protected operation is executed.

4. Reflection

Learning Achieved

Through my contribution, I learned how authentication and authorization are implemented in a REST API using Express.js and JWT. I understood the difference between authentication, which verifies the identity of a user, and authorization, which determines what an authenticated user is allowed to access.

I also learned how middleware can be used to protect API routes and how JWT tokens can carry information such as the user's role. Working on this module helped me understand why secrets such as the JWT secret and database connection details should be stored using environment variables instead of being directly written in the source code.

The testing process also improved my understanding of security validation. Testing an unauthorised request and receiving HTTP 401 confirmed that authentication checks must happen before protected business logic is executed. This gave me practical experience in designing a more secure and cloud-ready API.

Improvement Proposed

One improvement I would make is to strengthen the authentication system with refresh tokens and more detailed security controls such as account lockout or rate limiting for repeated failed login attempts. This would reduce the risk of token expiry affecting legitimate users while also providing better protection against repeated authentication attempts.

I would validate the improvement by testing successful login, access using an expired access token, refresh-token generation, invalid refresh tokens and repeated failed login attempts. I would also measure the response time of the authentication endpoints to ensure that the additional security controls do not introduce significant performance overhead.

5. Individual Declaration

Originality and Responsibility

I declare that the contribution, reflection and evidence presented in this document represent my own work and responsibility within Team 6. I personally participated in the design, implementation, testing and documentation of the Auth & Access Control module. Any external documentation or development tools used for understanding technologies such as Express.js and JWT were used only as supporting references. Any AI-assisted support used during development or documentation was reviewed and validated by me, and the final implementation and reflection are my responsibility.

Field	Student Entry
Student Signature	D.S.Tharuniya Priya
Date	02-09-2026

SUBMISSION CHECK: Typed Word document; personally written content; module evidence; working GitHub/commit links; similarity at or below 20%.